

Linear systems and elimination

Is the ideal GMRES bound sharp for every Jordan block?

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Historical assessment retained: challenging because the remaining Jordan-block min-max cases require control of multiple extremal singular vectors; specialist impact reflects a structural test case for GMRES theory.

Resolution — 2026-09-11

Solved affirmatively. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, proves the displayed equality for every $n \geq 2$, $1 \leq k < n$, and nonzero complex λ , with complex polynomials and starting vectors. No divisibility or eigenvalue-regime case remains open.

[Theorem 1 and its proof](#) establish the exact original target. The stronger Theorem 6 proves affine min-max equality for triangular Toeplitz matrices: finite Carathéodory–Fejér interpolation describes the maximal singular subspace, and scalar spectral factorization preserves every complex GMRES orthogonality equation in one unit vector. [Proof PDF](#) · [Standalone XeLaTeX source](#).

The complete AI-assisted proof passed a separate [independent Codex-agent review](#). This is automated-agent verification, without a claim of external human peer review or formal certification. [Submission record, preserved source and public eligibility check](#). The original problem statement and permanent ID remain unchanged; the earlier special cases and status checks below are retained as history.

Problem statement

For $n \geq 2$ and $\lambda \in \mathbb{C} \setminus \{0\}$, let $J_n(\lambda) = \lambda I + N$, where $N_{i,i+1} = 1$ and all other entries of N vanish. Let $\mathcal{P}_k = \{p \in \mathbb{C}[z] : \deg p \leq k, p(0) = 1\}$. Define

$$\psi_k(J) = \max_{\|v\|_2=1} \min_{p \in \mathcal{P}_k} \|p(J)v\|_2, \quad \phi_k(J) = \min_{p \in \mathcal{P}_k} \|p(J)\|_2.$$

Prove or disprove $\psi_k(J_n(\lambda)) = \phi_k(J_n(\lambda))$ for every $1 \leq k < n$. The maximum describes the slowest possible GMRES residual reduction, whereas the minimum over operator norms is the ideal bound. The familiar inequality $\psi_k \leq \phi_k$ does not answer the question. A single Jordan block is a published structural test case, not a claim about all nonnormal matrices.

References

Tichý, Liesen, and Faber, *On worst-case GMRES, ideal GMRES, and the polynomial numerical hull of a Jordan block*, ETNA 26 (2007), 453–473, §1 conjecture and subsequent special cases. Faber, Liesen, and Tichý, *Matrix best approximation in the spectral norm*, published in LAA 733 (2026), §§4–5.

Earlier status check — 2026-09-08

Searches for *Jordan block ideal GMRES equality proved 2026* and *site:arxiv.org GMRES Jordan block* found the original partial results and the 2026 general approximation paper, but no resolution of the displayed Jordan-block equality. The latter paper’s doubling theorem changes the matrix and does not by itself establish this statement.

Audit update — 2026-09-10

Rechecked the [author copy of Tichý–Liesen–Faber](#), especially §§3–5: Corollary 4.4 proves equality whenever k divides n , with additional eigenvalue regimes proved elsewhere in those sections. These are substantive parts of the displayed target. The [2025/2026 approximation paper](#) and targeted Jordan-block/ideal-GMRES searches did not supply the remaining cases; its doubling construction changes the input matrix.